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## Integrable Systems — Exercise Sheet 5

### *The Liouville–Arnold theorem*

#### Variant: Questions only

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Conventions, as in Lectures 2–5:  $\{f, g\} = \sum_i (\partial_{q_i} f \partial_{p_i} g - \partial_{p_i} f \partial_{q_i} g)$ ,  $\omega = \sum_i dq_i \wedge dp_i$ . The theorem's hypotheses are named as in the lecture:  $F_1, \dots, F_n$  in involution,  $\Sigma$  a connected component of a level set, (I) the  $dF_i$  independent at every point of  $\Sigma$ , (C)  $\Sigma$  compact.

**Exercise 1 (independence and involution).** (i) Let  $F$  be any smooth function on a phase space and  $g$  any smooth function of one variable. Show that  $\{F, g(F)\} = 0$  and that  $dF$  and  $d(g(F))$  are linearly dependent at every point. What does this say about counting functions in involution? (ii) On  $T^*\mathbb{R}^2$  with  $q \neq 0$ , let  $H = \frac{1}{2}|p|^2 + V(|q|)$  with  $V'(r) > 0$  for  $r > 0$ , and  $\ell = q_1 p_2 - q_2 p_1$ . Show that  $\{H, \ell\} = 0$ , and that  $dH$  and  $d\ell$  are linearly dependent exactly at the points where  $p$  is perpendicular to  $q$  with  $|p|^2 = |q| V'(|q|)$ : the circular orbits. (iii) Take the Kepler potential  $V = -1/r$ . For fixed  $\ell \neq 0$ , show that the smallest energy is  $-1/(2\ell^2)$ , reached on the circular orbit of radius  $\ell^2$ , and that the level set  $\{H = -1/(2\ell^2), \ell = \ell\}$  is exactly that circle. Which hypothesis of the theorem fails there, and which conclusion?

**Exercise 2 (deleting involution).** On  $\mathbb{R}^4$  let  $F_1 = \frac{1}{2}(q_1^2 + q_2^2 + p_1^2 + p_2^2)$  and  $F_2 = q_1$ , and fix  $c > 0$  and  $a$  with  $a^2 < 2c$ . (i) Show that  $F_1, F_2$  are not in involution. (ii) Show that the level set  $\Sigma = \{F_1 = c, F_2 = a\}$  is a 2-sphere, compact and connected, and that  $dF_1, dF_2$  are independent at every point of it. (iii) Show that  $\Sigma$  is not Lagrangian and not invariant under the flow of  $F_1$ . Which conclusion of the theorem fails, and why does this not contradict it?

**Exercise 3 (one degree of freedom, components, and a cylinder).** (i) (Stage (iv) with  $n = 1$ .) Let  $\Sigma$  be a compact connected component of a level set of  $H$  on which  $dH \neq 0$ , and  $x_0 \in \Sigma$ . The flow of  $H$  acts on  $\Sigma$ , and the lecture's stage (iii) shows that the orbit of  $x_0$  is all of  $\Sigma$ . Show that the stabiliser  $\Lambda = \{t : \varphi_t(x_0) = x_0\}$  is either  $\{0\}$  or  $\tau\mathbb{Z}$  for some  $\tau > 0$ , and that compactness excludes  $\{0\}$ . What is  $\Sigma$ , and what is  $\tau$ ? (ii) For  $H = \frac{1}{2}p^2 + \frac{1}{4}q^4 - \frac{1}{2}q^2$  at  $E = -\frac{1}{8}$  (Sheet 1, Exercise 4), how many components has the level set? What does the theorem say about the level set, and what about each component? (iii) The pendulum  $H = \frac{1}{2}p^2 - \cos \theta$  at  $E > 1$ , with  $\theta$  taken in  $\mathbb{R}$  (not modulo  $2\pi$ ): what is each component of the level set, and what is its stabiliser? Which hypothesis fails? What changes on the cylinder? (iv) On  $\mathbb{R}^4$  let  $F_1 = \frac{1}{2}p_1^2$  and  $F_2 = \frac{1}{2}(q_2^2 + p_2^2)$ . Show that they are in involution, and describe a component  $\Sigma$  of  $\{F_1 = e_1, F_2 = e_2\}$  for  $e_1, e_2 > 0$ . Find the stabiliser of the action of  $\mathbb{R}^2$  on it, and check the lecture's  $\Sigma \cong T^k \times \mathbb{R}^{n-k}$ .

**Exercise 4 (the two-dimensional oscillator, stages (ii)–(iv)).** Let  $H_i = \frac{1}{2}(p_i^2 + \nu_i^2 q_i^2)$  with  $\nu_i > 0$ ,  $i = 1, 2$ , and a point  $x_0$  with  $H_1(x_0), H_2(x_0) > 0$ . (i) Write the action of  $\mathbb{R}^2$ ,  $\Phi_{(t_1, t_2)}(x_0) = \varphi_{t_1}^{H_1} \circ \varphi_{t_2}^{H_2}(x_0)$ , explicitly. Why does the order of the two flows not matter? (ii) Find the stabiliser  $\Lambda$  of  $x_0$ , and deduce that the level set through  $x_0$  is a 2-torus. (iii) Take the basis  $e_1 = (2\pi/\nu_1, 0)$ ,  $e_2 = (0, 2\pi/\nu_2)$  of  $\Lambda$ . Find the angles  $\varphi = 2\pi A^{-1}t$  and the two basis cycles, and show that the flow of  $H = H_1 + H_2$  is  $\varphi(t) = \varphi(0) + (\nu_1, \nu_2)t$ .

**Exercise 5 ★.** *The spherical pendulum's energy–momentum map.* Away from the poles, with the conventions of optional lecture A (unit mass, length and gravity,  $\theta$  measured from the

upward vertical, so that the height is  $\cos \theta$ ),

$$H = \frac{1}{2} \left( p_\theta^2 + \frac{p_\phi^2}{\sin^2 \theta} \right) + \cos \theta, \quad J = p_\phi, \quad \theta \in (0, \pi).$$

- (i) Show that  $\{H, J\} = 0$ . With  $V_J(\theta) = J^2/(2 \sin^2 \theta) + \cos \theta$ , show that for  $J \neq 0$  the function  $V_J$  has exactly one critical point, a minimum at some  $\theta_J \in (\pi/2, \pi)$  with  $J^2 \cos \theta_J = -\sin^4 \theta_J$ .  
(ii) Deduce that for  $J \neq 0$  the energies reached are exactly  $E \geq E_{\min}(J) = V_J(\theta_J)$ , and show that  $E_{\min}(J) = -1 + |J| + O(J^2)$ . (iii) For  $J \neq 0$ , show that  $dH$  and  $dJ$  are dependent exactly on the circles  $p_\theta = 0$ ,  $\theta = \theta_J$  (the conical motions), and that for  $E > E_{\min}(J)$  the level set  $\{H = E, J = J\}$  is a 2-torus. (iv) At  $J = 0$  the pendulum swings in a vertical plane, and its two equilibria — hanging down and balanced upright — lie on the poles, outside these coordinates. Their energies are  $-1$  and  $1$ . Show that the value  $(J, E) = (0, 1)$  lies inside the image of the map  $(J, H)$ , above the curve  $E = E_{\min}(J)$ .